Lung Cancer in Bangladesh

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Background

Bangladesh is located in South Asia, sharing borders with India on the north, east, and west, Myanmar on the southeast, and the Bay of Bengal on the south (Fig. 1). It has a population of 166 million people, making it the world's eighth most populated country, with a density of 1265 individuals per square kilometer.[1] Most of the country's population (61%) lives in rural areas with a diverse cultural heritage influenced by Bengali, Islamic, and Hindu traditions. A variety of ethnic groups make up the country, including Bengali (98%), Bihari (0.3%), Chakma (0.3%), Meitei (0.1%), Khasi (0.1%), Santhal (0.1%), Oraon (0.1%), Munda (0.1%), and Rohingya (0.1%).[2] Furthermore, Bangladesh is home to some impressive natural landmarks, including the world's largest delta, the Sundarbans mangrove forest, and the picturesque hills of Chittagong and Sylhet.

Bangladesh is an emerging economy, with a strong focus on ready-made garments and agriculture. According to the World Bank, the country's gross domestic product per capita is $2457.9, and its gross domestic product (GDP) is growing at a rate of 6.9%.[3] The country spends 2.48% of its GDP and $45.86 per capita on health care. In terms of health care costs, the government contributes a low share, relying mainly on out-of-pocket payments from patients (73%).[4,5] Despite the challenges, Bangladesh has made significant strides in enhancing its health care system by improving infrastructure, human resources, and service delivery. As a result, the country has successfully lowered mortality and morbidity rates related to both communicable and noncommunicable diseases.[6] The current focus is on prioritizing urban primary health care for underserved urban populations through programs such as Community Clinics, which are initiatives aimed at improving health care access in these areas.[7] These clinics have become a flagship program of the government, making primary health care services accessible to rural populations and raising awareness about seeking formal health professionals instead of traditional healers.

Epidemiology

In Bangladesh, cancer is responsible for 10% of all annual deaths and ranks as the sixth leading cause of noncommunicable disease-related deaths.[8] The GLOBOCAN 2020 statistics report 156,775 new cancer cases and 108,990 cancer-related deaths in Bangladesh, with age-standardized rates of 106.2 and 75.3 per 100,000 for cancer incidence and mortality, respectively. Among them, lung cancer is the second most common in men (11.1% of new cases) and the sixth most common in women (4.6%) in Bangladesh.[9] Age-standardized incidence and mortality rates of cancer in Bangladesh as compared with neighboring (Southeast Asian) countries are depicted in Table 1.

Although cancer statistics for Bangladesh are primarily derived from the GLOBOCAN database, the country lacks a population-based cancer registry, making it difficult to determine the true incidence, prevalence,

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Figure 1. The location of Bangladesh with bordering countries.

and mortality estimates of cancer. The "Hospital Cancer Registry Report 2015-2017" by the National Institute of Cancer Research and Hospital reveals that lung cancer predominantly affects men (84.3%) compared with women, with a ratio of 5.4:1. The average age of patients is 58.85 years. The most common subtype of lung cancer is squamous cell carcinoma, accounting for 64.0% of the cases, and the report indicates an increase in its incidence in recent years.[10] Most patients were illiterate (66.2%) and belonged to a low socioeconomic group (55.4%), with a high prevalence of malnutrition (40.0%).[11]

Tobacco Use

A report titled "The Economic Cost of Tobacco Use in Bangladesh: A Health Cost Approach" has estimated that tobacco use is responsible for the deaths of 126,000 people in Bangladesh each year, which accounts for 13.5% of all deaths. Furthermore, approximately 1.5 million adults have diseases related to tobacco use and approximately 61,000 children are affected by second-hand smoke exposure.[12] In the fiscal year 2018-2019, direct health care costs accounted for nearly 9% of government health expenditures. Tobacco use in Bangladesh costs the economy $3.61 billion a year in productivity losses, equivalent to 1.4% of GDP in 2017 to 2018.[13] As per the Global Adult Tobacco Survey, 35.3% of adults use tobacco, and males are more likely to use it (46.0%) than females (25.2%).[14] The average age of tobacco initiation is 17.8 years, and 6.9% of youngsters aged 13 to 15 years smoke, with boys having a higher prevalence (9.2%) than girls (2.8%).[15] The government of Bangladesh has implemented several tobacco control

Table 1. Comparison of Age-Standardized Incidence Rate and Mortality Rate (ASR) of All Cancers and Lung Cancer in Bangladesh With Neighboring Countries (All ASRs Are Per 100,000 Persons)

Variables	Bangladesh	India	Pakistan	Myanmar	Nepal
Age-standardized incidence rate (all cancers)	106.2	97.1	110.4	136.8	80.9
Age-standardized mortality rate (all cancers)	75.3	63.1	74.3	99.0	54.8
Lung cancer, % (new cases)	8.3	5.5	5.9	11.0	12.2
Lung cancer rank by mortality	Second	Fourth	Fourth	First	First
5-year prevalence (all ages)/(per 100,000)	8.45	69.28	5.14	16.78	9.47
Age-standardized incidence rate (lung cancer)	9.5	5.4	7.0	15.7	10.4
Age-standardized mortality rate (lung cancer)	8.8	4.9	6.2	14.5	9.5

Source: GLOBOCAN 2020. The Global Cancer Observatory (https://gco.iarc.fr/today/data/factsheets/populations).
ASR, age-standardized rate.

measures since 2005, such as increasing tax on tobacco products, making public places smoke-free, imposing a comprehensive ban on tobacco advertising and promotion in national media, and informing people of the risks of tobacco use through warnings and information campaigns on tobacco packages.[16] The lack of resources and support for smoking cessation campaigns, such as quit helplines, clinics, and drugs, highlights the need for urgent action to enhance and expand smoking cessation efforts.

Screening and Early Detection

Studies have found the effectiveness of low-dose computed tomography (CT) screening in improving the survival rate of high-risk individuals with lung cancer in trials such as the National Lung Cancer Screening Trial and the Dutch-Belgian NELSON trial.[17,18] Nevertheless, Bangladesh currently lacks a formal lung cancer screening program. Despite the availability of low-dose CT in Bangladesh, screening is only accessible through out-of-pocket payment for high-risk individuals. The absence of screening programs, limited access to health care facilities in rural areas, delayed consultations with physicians after symptom onset, use of antitubercular regimens due to the high prevalence of pulmonary tuberculosis, and delays in conducting diagnostic tests such as CT scans and biopsy may all contribute to late-stage diagnosis of lung cancer. To prevent lung cancer, it is crucial to establish an integrated program for tobacco control and early disease detection.

Diagnosis and Staging

Obtaining an accurate diagnosis and staging of lung cancer is crucial for effective treatment planning, outcome prediction, and disease monitoring. Biopsy is the standard method for confirming a histologic diagnosis, typically performed through bronchoscopy or transthoracic sampling in specialized centers in Bangladesh. Nevertheless, due to limited resources and financial constraints, many centers in the country resort to fine-needle aspiration cytology for diagnosis, which can present additional challenges for clinicians. In Bangladesh, endobronchial ultrasound-guided transbronchial needle aspiration is available in tertiary academic centers, whereas transthoracic sampling under image guidance is performed by interventional radiologists but is only accessible in limited tertiary centers. Immunohistochemistry services are primarily offered by academic institutions and private laboratories, but their availability is limited due to high costs, with most located in the capital city. To reduce costs, patients with advanced nonsquamous cell carcinoma or those without a smoking history or with mixed histological types are typically advised to undergo testing for *EGFR*, *ALK*, and *ROS-1* mutations. Approximately 23% of Bangladesh's population is affected by the most common *EGFR* mutations, which are exons 19 and 21, similar to other Asian studies.[19] Nevertheless, the availability of next-generation sequencing testing is limited due to its high cost, and delays in diagnosis can occur when outsourcing services to other countries. Imaging services such as magnetic resonance imaging, CT scans, and positron emission tomography-CT scans are concentrated in urban areas, leading to further delays in public facilities. Table 2 reveals the health system facilities for every 10,000 patients with cancer in Bangladesh.[20] The Bangladeshi government is currently in the process of establishing comprehensive cancer care centers in each of the eight divisions with 180 beds to provide proper diagnosis and decentralized care for patients with cancer.

Education

In Bangladesh, the Bangabandhu Sheikh Mujib Medical University, a state-sponsored university, is the only teaching institution that provides postgraduate degree in medical oncology, surgical oncology, and radiation oncology. After completing M.B.B.S. degree, students can

Table 2. Health System Capacity for Patients With Cancer in Bangladesh From 2019 to 2020

Facilities	Total	For Every 10,000 Patients With Cancer
Cancer centers/dedicated cancer ward	29	1.85 (2020 GLOBOCAN)
External beam radiotherapy	31	1.98 (2020 GLOBOCAN)
Mammographs	149	9.9 (2018 GLOBOCAN)
CT scanners	183	12.1 (2018 GLOBOCAN)
MRI scanners	70	4.6 (2018 GLOBOCAN)
PET or PET/CT scanners	7	0.50 (2020 GLOBOCAN)
Palliative care availability		Yes
Geriatric oncology		Not available
Population-based cancer registry		Not available
Dedicated lung cancer center		Not available

Source: WHO. Cancer Bangladesh 2020 Country Profile, The Global Cancer Observatory—Bangladesh, March 2021. DGHS. Health Bulletin 2020 Bangladesh: DGHS; 2023.

CT, computed tomography; DGHS, Director General of Health Services; MRI, magnetic resonance imaging; PET, positron emission tomography.

Table 3. Health System of Workforce for Patients With Cancer in Bangladesh 2019 to 2020

Workforces	Total	For Every 10,000 Patients With Cancer
Radiation oncologist	203	12.95 (2020 GLOBOCAN)
Medical oncologist	30	1.91 (2020 GLOBOCAN)
Chest surgeon	30	1.91 (2020 GLOBOCAN)
Medical physicist	331	4.02 (2020 GLOBOCAN)
Radiologist	501	33.2 (2019 GLOBOCAN)
Nuclear medicine physician	100	6.6 (2019 GLOBOCAN)

Source: WHO. Cancer Bangladesh 2020 Country Profile, Director General of Health Services. Health Bulletin 2020 Bangladesh: DGHS; 2023, Bangladesh Society of Radiation Oncologist, Medical Oncology Society in Bangladesh.
DGHS, Director General of Health Services.

apply for a five-year residency program that leads to specialization as an "Oncologist" in Medical Oncology, Surgical Oncology, or Radiation Oncology. Nevertheless, advanced training in medical or radiation oncology is not currently available in the country. In addition, The Bangladesh College of Physicians and Surgeons offers a five-year fellowship program in radiotherapy (RT) and medical oncology. According to the WHO Cancer Country Profile 2020, Bangladesh has only 1.7 radiation oncologists, 21.9 medical physicists, and 173.4 surgeons per 10,000 patients with cancer.[20] As of now, there are only approximately 30 medical oncologists in the country for the population of approximately 177 million.

Treatment

Oncology services in Bangladesh are recommended to hold a multidisciplinary team (MDT) meeting before treating patients with cancer, but the recommendations are not consistently followed due to a lack of resources (Table 3). As Bangladesh does not have nationally standardized lung cancer treatment guidelines, most oncologists use international guidelines (e.g., National Comprehensive Cancer Network, European Society of Medical Oncology, American Society of Clinical Oncology) or consensus strategies developed in MDT meetings to determine patient management plans. To maintain a standard treatment protocol throughout the country, a few professional organizations, such as the Medical Oncology Society in Bangladesh, the Bangladesh Association for Thoracic Surgery, the Bangladesh Cancer Society, and the Oncology Club, have organized workshops and seminars with international experts and local lung cancer specialists. Each month, the group discusses difficult cases, promotes lung cancer research in Bangladesh, and initiates the development of nationally standardized lung cancer guidelines.

Thoracic Surgery

The characteristics of patients diagnosed with having lung cancer during their initial visit to a public tertiary-level hospital in Bangladesh have been provided in Table 4. Surgery is the main curative treatment for early stage lung cancer.[21] Currently, surgical treatment for lung cancer is only available in two Metropolitan cities, Dhaka and Chittagong, where at least two public and eight private thoracic surgery centers perform lung cancer resections using both open thoracotomies and video-assisted thoracoscopic surgery. Robotic surgery is, however, not yet available in public or private hospitals in Bangladesh. On the basis of the pathologic findings, patients with stages I to II NSCLC undergo surgical resection with mediastinal lymph node sampling followed by adjuvant systemic therapy. An MDT usually evaluates patients with stage III NSCLC. Neoadjuvant

Table 4. The Characteristics of Patients With Lung Cancer in a Public Tertiary-Level Hospital in Bangladesh (at First Visit)

Characteristics	Men (n = 1580)	Women (n = 288)	Total (N = 1868)
BMI (kg/m^2)	19.5 ± 3.3	20.7 ± 4.5	19.7 ± 3.6
Illiterate (no formal schooling), n (%)	1016 (65)	210 (74)	1227 (66)
Low socioeconomic status, n (%)	866 (55)	159 (56)	1025 (55)
Any form of tobacco use, n (%)	1451 (91.8)	216 (75)	1667 (89.2)
Visceral metastasis, n (%)	146 (19.6)	29 (18.1)	175 (19.4)
Comorbidity, n (%)	444 (28)	107 (37)	551 (29.5)
ECOG $\geq$ 2, n (%)	1042 (66.1)	182 (63.06)	1224 (65.6)
Best supportive care, n (%)	634 (40.1)	114 (39.6)	748 (40)
Death within 1 y after starting the treatment, n (%)	1235 (78.2)	234 (81.3)	1470 (78.7)

Source: Department of Medical Oncology, National Institute of Cancer Research and Hospital, Dhaka, Bangladesh.
BMI, body mass index; ECOG, Eastern Cooperative Oncology Group.

systemic therapy and surgical resection are applied if the patient has potentially resectable disease. Usually, RT with concurrent chemotherapy is recommended when a patient's cancer is deemed unresectable. To minimize postoperative and intraoperative complications, surgeons typically avoid perioperative RT. Performing thoracic surgery in Bangladesh comes with certain challenges. These include the high cost of the procedures and patient-related factors such as smoking, poor nutritional status, and reduced lung capacity. In addition, the limited number of skilled thoracic surgeons and the high prevalence of tuberculosis and other infectious diseases can lead to dense pleural adhesions, making surgery more difficult. Despite these challenges, the mortality rate associated with thoracic surgery in Bangladesh is relatively low, at approximately 3.3%.[22]

Table 5. Antineoplastic Agents for Lung Cancer Available in Bangladesh

Chemotherapy	Targeted Therapy	Immunotherapy
Cisplatin	Afatinib	Pembrolizumab
Carboplatin	Alectinib	Nivolumab
Cyclophosphamide	Brigatinib	
Docetaxel	Crizotinib	
Doxorubicin/ epirubicin	Erlotinib	
Etoposide	Gefitinib	
Gemcitabine	Osimertinib	
Irinotecan		
Paclitaxel		
Pemetrexed		
Vinorelbine		

Radiation Therapy

RT is a treatment option for lung cancer at all stages, including postoperative RT for positive surgical margins, definitive RT for early stage disease, concurrent or sequential chemo-RT for locally advanced disease, and palliative RT for metastatic disease. According to the WHO, Bangladesh needs approximately 160 to 180 RT centers with at least 180 to 200 RT machines to provide

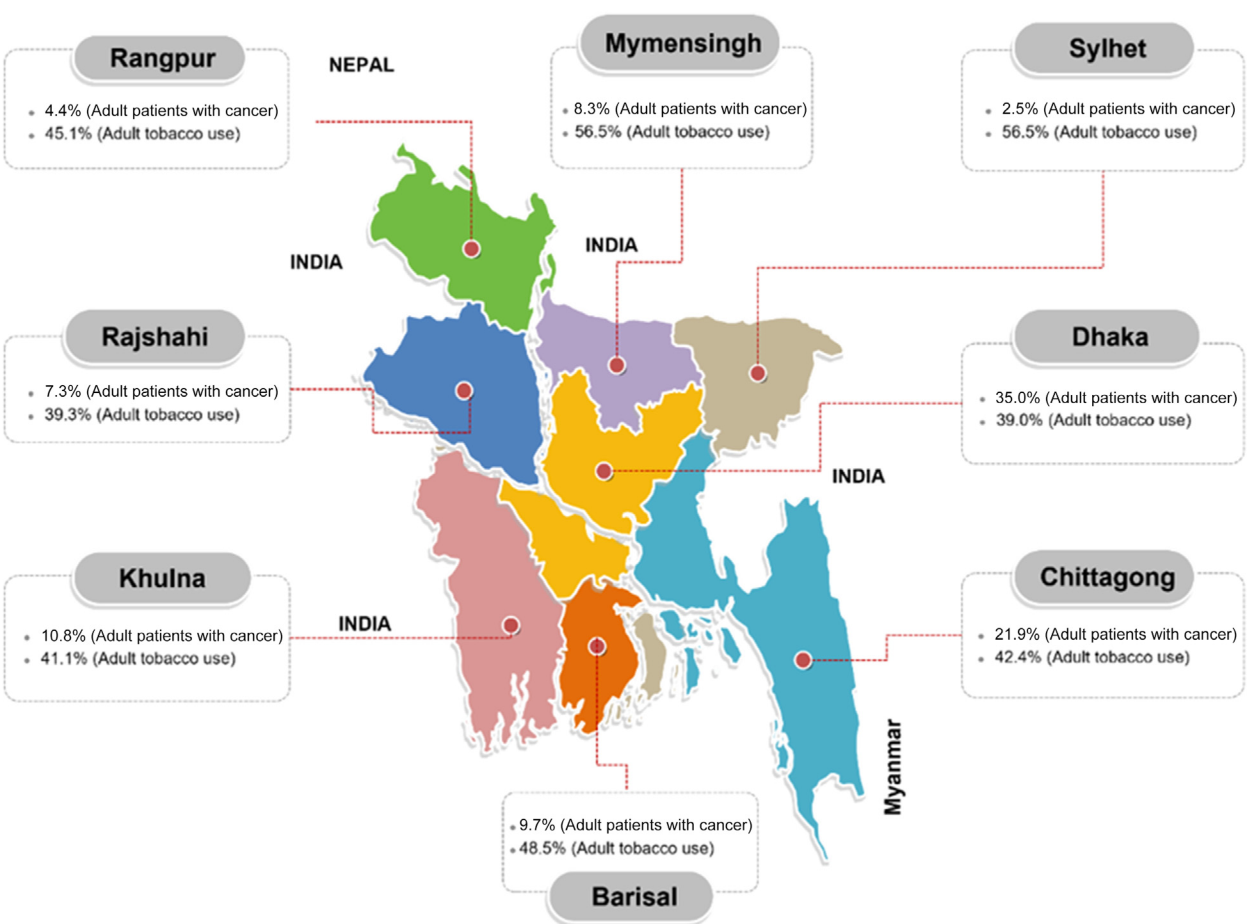

Figure 2. Map of Bangladesh, cancer patient distribution division wise: Source: "Hospital Cancer Registry Report 2015-2017 (NICRH)," Variation in tobacco use amongst all adults, aged 18 to 69 years, by division, Bangladesh STEPs survey, 2018(21),[24] Location of proposed 8 divisional comprehensive cancer care center.

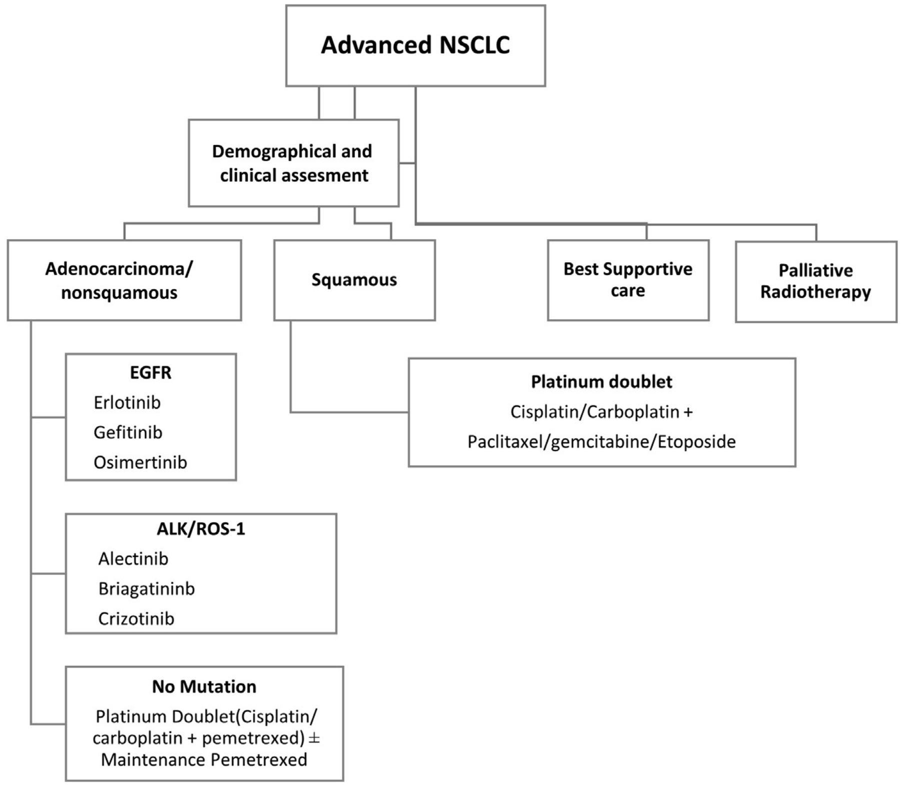

Figure 3. Flowchart of systemic therapy for advanced NSCLC in a public hospital of Bangladesh. Usually, immunotherapy is not available in public hospital.

adequate care.[23] The country currently has 15 government-operated centers and 11 nongovernment-operated centers. Only a few government centers (five Linacs and three Cobalts) have Linacs and Cobalt machines that perform three-dimensional conformal radiation therapy (3DCRT), intensity-modulated radiation therapy, and conventional two-dimensional RT. Meanwhile, nongovernment centers have 18 Linacs and three Cobalt machines that perform 3DCRT, intensity-modulated radiation therapy, two-dimensional traditional planning, and stereotactic body RT and stereotactic RT using four-dimensional CT planning. A 3DCRT and CT-based planning approach is most often used in functional centers to treat lung cancer. Whenever the planning target volume cannot be adequately covered while meeting the dose constraints for the organs at risk, intensity-modulated RT is used. To verify image-guided treatment, orthogonal electronic portal imaging devices and cone-beam imaging are used when available. Standard RT involves delivering 60 Gy in 30 fractions in 6 weeks, with 2 Gy fractions daily. Nevertheless, hypofractionation schedules, such as 55 Gy in 20 fractions, 40 Gy in 15 fractions, 30 Gy in 10 fractions, or 20 Gy in five fractions, may be used on the basis of the treatment intent and the patient's general health condition. For early stage lung cancers (T1 to T2, N0M0), stereotactic body RT with a dose of more than 100 Gys BED is the preferred treatment, but only three centers offer it. Because of the prolonged wait times in public facilities (approximately 2–5 mo) for RT, patients have had negative disease outcomes, and the need to seek treatment in private facilities for prompt results has resulted in financial toxicity.

Systemic Therapy

The introduction of targeted therapy and immunotherapy has led to significant changes in the systemic treatment of NSCLC, and the oncological practice in Bangladesh now has access to systemic therapies listed in Table 5. In Bangladesh, platinum doublet therapy is still the standard of care for advanced NSCLC, but drug selection depends on the patient's characteristics and tumor subtype (Fig. 2). Patients with lung cancer often receive chemotherapeutic agents such as cisplatin, carboplatin, pemetrexed, taxanes (paclitaxel, docetaxel), gemcitabine, etoposide, and vinorelbine. Patients with EGFR-mutant NSCLC are typically treated with tyrosine kinase inhibitors such as gefitinib, erlotinib, and osimertinib, whereas patients with *ALK* and *ROS-1*–rearranged NSCLC are treated with crizotinib. In SCLC, a platinum-etoposide doublet is the mainstay of systemic therapy, with cisplatin-etoposide often used for limited-stage

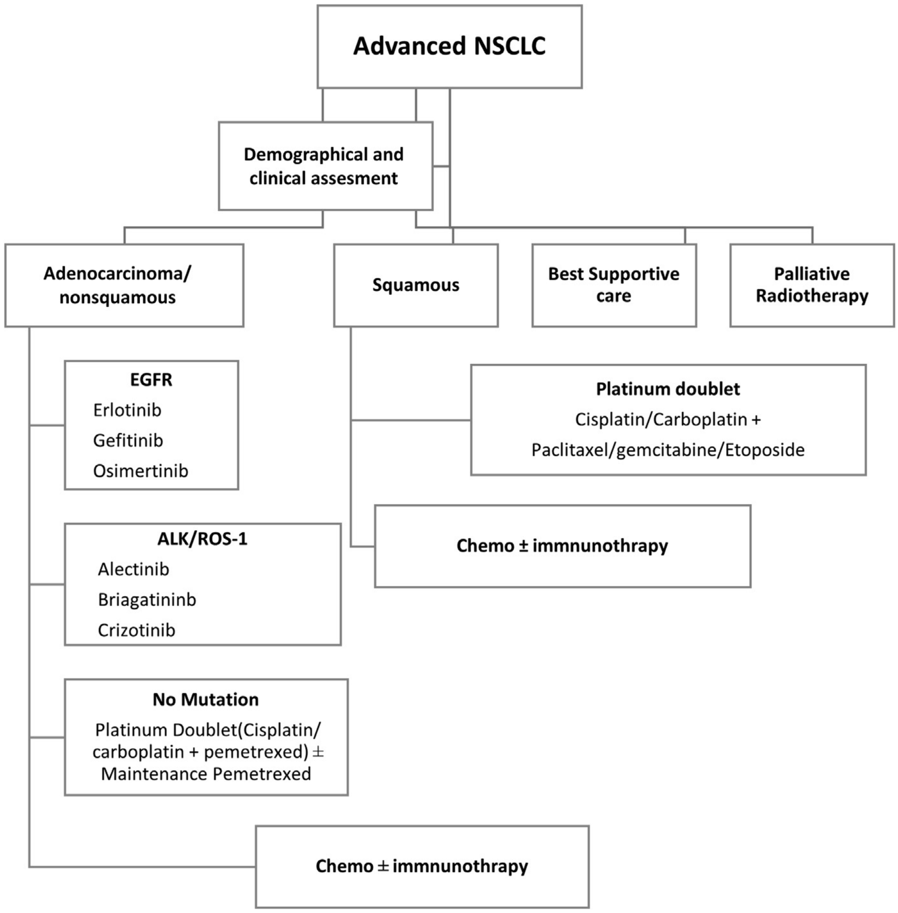

Figure 4. Flowchart of systemic therapy for advanced NSCLC in a private hospital of Bangladesh. Chemo, chemotherapy.

disease and given concurrently with radiation. In common practice, carboplatin is substituted for cisplatin in extensive-stage disease, for older frail patients, and when the potential toxicity from cisplatin use is a concern. In the second-line setting, irinotecan, paclitaxel, and CAV (cyclophosphamide, adriamycin, vincristine) regimens are often used. Figures 3 and 4 provides a summary of the treatment of NSCLC in public and private settings.

Pembrolizumab or nivolumab, immune checkpoint inhibitors, have been available in Bangladesh since 2020, but their high cost prevents widespread use. There are currently 15 public and 13 private centers in Bangladesh that provide systemic anticancer therapy, with 13 of these centers located in the capital city of Dhaka. Molecular diagnostic testing is also available in both public and private centers. Nevertheless, access to these facilities is limited, and a limited panel of targetable mutations for the available drugs is recommended for practical patient management to reduce treatment costs. The Bangladesh government provides most chemotherapy drugs and some targeted therapies, including gefitinib, erlotinib, bevacizumab, and trastuzumab, free of charge through public hospitals.

Lung Cancer Treatment in Geriatric Patients

The elderly population in Bangladesh is growing at a rate 3.41 times faster than the overall population growth and currently constitutes 9.28% of the total population.[25] Because geriatric patients often have unique needs and challenges, comprehensive geriatric assessments are essential to ensure the best possible outcomes for patients with lung cancer. Elderly patients may experience more adverse effects from chemotherapy or radiation therapy and may benefit more from less aggressive treatments.[26] A multidisciplinary approach involving oncologists, geriatricians, nurses, and other health care professionals can provide the best care for elderly patients with lung cancer. Unfortunately, Bangladesh lacks specialized centers or postgraduate courses for geriatric oncology. Nevertheless, there are some initiatives such as Geriatric Medicine unit in Dhaka Medical College (2014), the Parents Care Act 2013, National Policy on Ageing in 2007, Monthly Geriatric allowance, six old-age homes, and nongovernmental organizations providing care for the elderly.[27] Caregivers may also require significant support, especially if they are elderly or have health issues. To enhance the quality of geriatric oncology care in Bangladesh, it is crucial to establish specialized centers focusing on geriatric oncology and to train more health care professionals in this field. It is also necessary to develop guidelines for managing elderly patients with lung cancer, increase access to palliative care, and raise public awareness about geriatric oncology through educational campaigns. Furthermore, more funding from both government and private sectors is required to support research, infrastructure development, and training programs.

Conclusions

Lung cancer is a significant public health concern in Bangladesh, and effective tobacco control measures and smoking cessation programs are needed to reduce morbidity and mortality. Lack of awareness contributes to delayed diagnosis, highlighting the need for health education and context-specific screening programs for physicians and patients. The development of a national lung cancer diagnostic and treatment guideline is crucial to ensuring high-quality care. Improving cancer care requires a focus on standard medical oncology, pathological, radiology and imaging, surgery, RT, and supportive facilities, including training more oncologists throughout the country. There is a lack of local data on lung cancer epidemiology and genetics, highlighting the urgent need for a population-based cancer registry system. Global cancer research must be expanded, particularly in Bangladesh, and collaboration with institutions in developed countries can be valuable in treatment, patient advocacy, and research.

CRediT Authorship Contribution Statement

Muhammad Rafiqul Islam: Conceptualization, Methodology, Visualization, Writing—original draft preparation, Writing—review and editing.

Syeda Masuma Siddiqua: Project administration, Data curation, Writing—review and editing.

Rashedul Islam: Data curation, Visualization.

Anwar Hossain: Investigation, Data curation.

Salman Bashar Al Ayub: Project administration, Writing—review and editing.

Md. Shariful Islam: Data curation, Investigation.

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Nazrina Khatun: Supervision, Validation.

Md Nazmul Karim: Conceptualization, Supervision, Writing—reviewing and editing.

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